# Permutations & Combinations

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**ANSWER KEY — COMPLETE WORKED SOLUTIONS**

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** **[Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Find the center, radius, and standard Cartesian equation of the circle given by the general equation $x^2 + y^2 - 8x + 6y - 11 = 0$.

Step 1: Group variables and complete the square: $(x^2 - 8x) + (y^2 + 6y) = 11$. $(x^2 - 8x + 16) + (y^2 + 6y + 9) = 11 + 16 + 9 = 36$. $(x - 4)^2 + (y + 3)^2 = 36$. Step 2: Extract circle invariants: Center $(h, k) = (4, -3)$ and radius $r = \sqrt{36} = 6$.

**2** **[Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

In triangle $ABC$, side $a = 7$ cm, side $b = 9$ cm, and angle $A = 42^\circ$. (a) Show that two distinct triangles can be formed with these given measurements. (b) Calculate both possible values for angle $B$ and the corresponding values of side $c$.

(a) Height of triangle from vertex $C$: $h = b\sin A = 9\sin 42^\circ \approx 9(0.6691) = 6.022$ cm. Since $h < a < b$ (i.e. $6.022 < 7 < 9$), the side $a$ is long enough to reach the base line on either side of the altitude. Therefore, exactly two non-congruent triangles exist. ■ (b) Sine rule: $\frac{\sin B}{b} = \frac{\sin A}{a} \Rightarrow \sin B = \frac{9\sin 42^\circ}{7} \approx \frac{6.0221}{7} \approx 0.8603$. First triangle (acute): $B_1 = \arcsin(0.8603) \approx 59.35^\circ \approx 59.4^\circ$..

**3** **[Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Three mobile antennas are located at coordinates $A(0, 0)$, $B(6, 0)$, and $C(2, 4)$. (a) Find the equation of the perpendicular bisector of segment $AB$. (b) Find the equation of the perpendicular bisector of segment $AC$. (c) Find the coordinates of the Voronoi vertex (circumcentre) equidistant from all three antennas, and calculate the radius of the circumcircle.

(a) For segment $AB$: $A(0, 0)$ and $B(6, 0)$ lie on the $x$-axis. Midpoint $M_{AB} = (3, 0)$. The perpendicular bisector is the vertical line $x = 3$. (b) For segment $AC$: $A(0, 0)$ and $C(2, 4)$. Midpoint $M_{AC} = \left(\frac{0+2}{2}, \frac{0+4}{2}\right) = (1, 2)$. Gradient $m_{AC} = \frac{4-0}{2-0} = 2$. Perpendicular gradient $m_\perp = -\frac{1}{2}$. Equation: $y - 2 = -\frac{1}{2}(x - 1) \Rightarrow y = -\frac{1}{2}x + \frac{5}{2} \Leftrightarrow x + 2y = 5$. (c) The Voronoi vertex is the intersectio.

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**4** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Six wines were evaluated by two independent sommeliers, $X$ and $Y$, giving the following ratings: Wine: A, B, C, D, E, F Judge X: 88, 72, 95, 60, 81, 75 Judge Y: 84, 78, 91, 65, 85, 70 (a) Construct a table of ranks for both judges. (b) Calculate Spearman's rank correlation coefficient $r_s$. (c) State what this coefficient indicates about the judges' agreement.

(a) Ranking in descending order (highest rating = rank 1): Judge X: 95 (1), 88 (2), 81 (3), 75 (4), 72 (5), 60 (6). Judge Y: 91 (1), 85 (2), 84 (3), 78 (4), 70 (5), 65 (6). Table of paired ranks and differences $d_i = R_X - R_Y$: Wine A: $R_X = 2, R_Y = 3 \Rightarrow d_1 = -1, d_1^2 = 1$ Wine B: $R_X = 5, R_Y = 4 \Rightarrow d_2 = 1, d_2^2 = 1$ Wine C: $R_X = 1, R_Y = 1 \Rightarrow d_3 = 0, d_3^2 = 0$ Wine D: $R_X = 6, R_Y = 6 \Rightarrow d_4 = 0, d_4^2 = 0$ Wine E: $R_X = 3, R_Y = 2 \Rightarrow d_5 = 1, d_5^2 = 1$.

**5** [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A small business borrows USD 40,000 to renovate its premises. The loan carries an annual nominal interest rate of 7.2% compounded monthly, with equal monthly repayments made at the end of each month over 6 years. (a) Calculate the monthly repayment amount. (b) Calculate the total amount repaid and the total interest charged on the loan. (c) After making payments for 3 full years, determine the outstanding loan balance.

(a) Total periods $N = 6 \times 12 = 72$. Monthly interest rate $r = \frac{7.2\%}{12} = 0.6\% = 0.006$. Using the present value of an annuity formula: $PV = PMT\left[\frac{1-(1+r)^{-N}}{r}\right] \Rightarrow 40,000 = PMT\left[\frac{1-(1.006)^{-72}}{0.006}\right]$. $(1.006)^{-72} \approx 0.65028 \Rightarrow 1 - 0.65028 = 0.34972$. Fraction: $\frac{0.34972}{0.006} \approx 58.2866$. $PMT = \frac{40,000}{58.2866} \approx USD686.26$. (b) Total amount repaid: $72 \times 686.264 = USD49,411.01$. Total intere.

**6** [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A curve has equation $f(x) = \frac{2x^2}{x^2+k}$ for some positive constant $k > 0$. (a) Write down the horizontal asymptote of $y = f(x)$. (b) Find $f'(x)$ in terms of $k$. (c) Given that the gradient of the tangent to the curve at $x = 1$ is equal to $\frac{1}{2}$, find all exact possible values of $k$.

(a) $\lim_{x\to\pm\infty} \frac{2x^2}{x^2+k} = \lim_{x\to\pm\infty} \frac{2}{1+k/x^2} = 2$. Horizontal asymptote: $y = 2$. (b) Apply the quotient rule with $u = 2x^2$ and $v = x^2 + k$: $u' = 4x$, $v' = 2x$. $f'(x) = \frac{4x(x^2+k) - 2x^2(2x)}{(x^2+k)^2} = \frac{4x^3 + 4kx - 4x^3}{(x^2+k)^2} = \frac{4kx}{(x^2+k)^2}$. (c) At $x = 1$, $f'(1) = \frac{1}{2}$: $\frac{4k(1)}{(1^2+k)^2} = \frac{1}{2} \Rightarrow \frac{4k}{(k+1)^2} = \frac{1}{2} \Rightarrow 8k = (k+1)^2$. Expand.

**7** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find all distinct points of intersection between the two polar curves $r = 2\sin\theta$ and $r = 2\cos\theta$ on $0 \le \theta < 2\pi$.

Equate the two equations: $2\sin\theta = 2\cos\theta \Rightarrow \tan\theta = 1$. On $[0, 2\pi)$, $\theta = \frac{\pi}{4}$ or $\theta = \frac{5\pi}{4}$.For $\theta = \frac{\pi}{4}$: $r = 2\sin\left(\frac{\pi}{4}\right) = \sqrt{2}$, giving point $\left(\sqrt{2}, \frac{\pi}{4}\right)$.For $\theta = \frac{5\pi}{4}$: $r = 2\sin\left(\frac{5\pi}{4}\right) = -\sqrt{2}$. However, $\left(-\sqrt{2}, \frac{5\pi}{4}\right)$ represents the identical geometric point as $\left(\sqrt{2}, \frac{\pi}{4}\right)$.Next, check the pole.

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## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

A closed cylindrical metal container with circular base of radius $r$ cm and height $h$ cm has a fixed volume of $250\pi$ cm$^3$. (a) Express $h$ in terms of $r$. (b) Show that the total surface area is $A(r) = 2\pi r^2 + \frac{500\pi}{r}$. (c) Find the radius $r$ that minimizes the surface area, and show that for this optimal cylinder, the height equals the diameter ($h = 2r$).

(a) Volume of cylinder: $V = \pi r^2 h = 250\pi \Rightarrow h = \frac{250}{r^2}$. (b) Total surface area includes 2 circular ends plus curved surface: $A = 2\pi r^2 + 2\pi r h = 2\pi r^2 + 2\pi r\left(\frac{250}{r^2}\right) = 2\pi r^2 + \frac{500\pi}{r}$. (c) Differentiate with respect to $r$: $A'(r) = 4\pi r - \frac{500\pi}{r^2}$. Set $A'(r) = 0$: $4\pi r = \frac{500\pi}{r^2} \Rightarrow 4r^3 = 500 \Rightarrow r^3 = 125 \Rightarrow r = 5$ cm. Check second derivative.

**9** **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

A continuous random variable $X$ is normally distributed with mean $\mu$ and standard deviation $\sigma$. Given that $P(X < 45) = 0.12$ and $P(X > 85) = 0.08$, find the value of $\mu$ and $\sigma$ correct to 3 significant figures.

1. Standardize using $Z = \frac{X - \mu}{\sigma}$: For the lower tail: $P(X < 45) = 0.12 \Rightarrow P(Z < z_1) = 0.12$. Using GDC `invNorm(0.12, 0, 1)`: $z_1 = -1.17499$. For the upper tail: $P(X > 85) = 0.08 \Rightarrow P(X < 85) = 0.92 \Rightarrow P(Z < z_2) = 0.92$. Using GDC `invNorm(0.92, 0, 1)`: $z_2 = 1.40507$. 2. Form the system of equations: $\frac{45 - \mu}{\sigma} = -1.1750 \Rightarrow \mu - 1.1750\sigma = 45$. $\frac{85 - \mu}{\sigma} = 1.4051 \Rightarrow \mu + 1.4051\sigma = 85$. 3. Subtract the first.

**10** **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

Consider the function $f(x) = x^2 e^{-x}$. (a) Find $f'(x)$ and express it in factored form. (b) Hence, find the coordinates of all stationary points of $f$. (c) Use the second derivative test to classify each stationary point as a local maximum or local minimum.

(a) Apply the product rule with $u = x^2$ and $v = e^{-x}$: $u' = 2x$, $v' = -e^{-x}$. $f'(x) = 2xe^{-x} + x^2(-e^{-x}) = (2x - x^2)e^{-x} = x(2 - x)e^{-x}$. (b) Set $f'(x) = 0$. Since $e^{-x} > 0$ for all $x$: $x(2 - x) = 0 \Rightarrow x = 0 \quad \text{or} \quad x = 2$. Coordinates: At $x = 0$: $f(0) = 0^2 e^0 = 0 \Rightarrow (0, 0)$. At $x = 2$: $f(2) = 2^2 e^{-2} = 4e^{-2} = \frac{4}{e^2} \approx 0.5413 \Rightarrow \left(2, \frac{4}{e^2}\right)$. (c) Compute $f''(x)$ by differentiating $(2x - x^2)e^{-x}$.

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**11** **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Prove by direct deductive reasoning that the difference between the squares of any two odd integers is always divisible by 8.

1. Let $m$ and $n$ be any two odd integers. Then they can be expressed algebraically as $m = 2k + 1$ and $n = 2j + 1$ for some integers $k, j \in \mathbb{Z}$. 2. Form the difference of their squares: $m^2 - n^2 = (2k+1)^2 - (2j+1)^2 = (4k^2 + 4k + 1) - (4j^2 + 4j + 1) = 4k^2 + 4k - 4j^2 - 4j = 4[k(k+1) - j(j+1)]$. 3. Analyze the products $k(k+1)$ and $j(j+1)$: For any integer $k$, exactly one of $k$ or $k+1$ is even. Therefore, the product of any two consecutive integers is always divisible by 2.

**12** **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

Determine whether the lines $\mathbf{r}_1 = (1; 2; -1) + \lambda(2; 1; 3)$ and $\mathbf{r}_2 = (0; 5; 4) + \mu(-1; 2; 1)$ intersect or are skew.

1. Check parallelism: Direction vectors are $\mathbf{d}_1 = (2; 1; 3)$ and $\mathbf{d}_2 = (-1; 2; 1)$. Since $\frac{2}{-1} eq \frac{1}{2}$, they are not scalar multiples, so the lines are not parallel. 2. Equate components: $1 + 2\lambda = -\mu \Rightarrow 2\lambda + \mu = -1 \quad (1)$ $2 + \lambda = 5 + 2\mu \Rightarrow \lambda - 2\mu = 3 \quad (2)$ $-1 + 3\lambda = 4 + \mu \Rightarrow 3\lambda - \mu = 5 \quad (3)$ Add (1) and (3): $5\lambda = 4 \Rightarrow \lambda = \frac{4}{5}$. From.

**13** **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Decompose the vector $\mathbf{u} = \langle 6, 2 \rangle$ into components parallel and orthogonal to $\mathbf{v} = \langle 3, 4 \rangle$.

Step 1: Compute dot product and squared magnitude: $\mathbf{u} \cdot \mathbf{v} = (6)(3) + (2)(4) = 18 + 8 = 26$. $||\mathbf{v}||^2 = 3^2 + 4^2 = 9 + 16 = 25$. Step 2: Vector projection (parallel component): $\mathbf{u}_{\parallel} = \text{proj}_{\mathbf{v}}\mathbf{u} = \left(\frac{\mathbf{u} \cdot \mathbf{v}}{||\mathbf{v}||^2}\right)\mathbf{v} = \frac{26}{25}\langle 3, 4 \rangle = \left\langle \frac{78}{25}, \frac{104}{25} \right\rangle = \langle 3.12, 4.16 \rangle$. Step 3: Orthogonal component.

**14** **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Find the exact area of the finite region enclosed between the parabola $y = 3x - x^2$ and the line $y = x - 3$.

1. Find intersection points: $3x - x^2 = x - 3 \Rightarrow x^2 - 2x - 3 = 0 \Rightarrow (x-3)(x+1) = 0 \Rightarrow x = -1 \text{ and } x = 3.$ 2. On the interval $[-1, 3]$, the parabola is concave down and lies above the line: At $x = 0$: $y_{\text{parabola}} = 0 > -3 = y_{\text{line}}$. 3. Set up the definite integral: $A = \int_{-1}^{3} [(3x - x^2) - (x - 3)]\,dx = \int_{-1}^{3} (3 + 2x - x^2)\,dx$. 4. Integrate.

## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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**15** **[Challenging]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)** • **Cross-Lesson: 12.01 Forces & Equilibrium**

Practice AI.1: Use the trapezoidal rule with 4 subintervals to estimate $\int_1^3 \sqrt{1+x^3}\,dx$.

1. Subinterval width: $h = \frac{b-a}{n} = \frac{3-1}{4} = 0.5$. 2. Evaluation grid points:
$x_0 = 1, x_1 = 1.5, x_2 = 2.0, x_3 = 2.5, x_4 = 3.0$. 3. Compute function values $y_i = \sqrt{1+x_i^3}$:
$y_0 = \sqrt{1+1} = \sqrt{2} \approx 1.4142$ $y_1 = \sqrt{1+1.5^3} = \sqrt{1+3.375} = \sqrt{4.375} \approx 2.0917$
$y_2 = \sqrt{1+2^3} = \sqrt{9} = 3.0000$ $y_3 = \sqrt{1+2.5^3} = \sqrt{1+15.625} = \sqrt{16.625} \approx 4.0774$
$y_4 = \sqrt{1+3^3} = \sqrt{28} \approx 5.2915$ 4. Apply the trapezoidal r.

**16** **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)** • **Cross-Lesson: 13.01 Newton's Laws**

Consider the points $A(1,2,3)$, $B(2,4,1)$, and $C(-1,0,5)$ in $\mathbb{R}^3$. Find a non-zero vector $\mathbf{n}$ normal to the plane containing $A, B, C$, and write the Cartesian equation of the plane.

Step 1: Form vectors $\vec{AB}$ and $\vec{AC}$: $\vec{AB} = \langle 2-1, 4-2, 1-3\rangle = \langle 1, 2, -2\rangle$.
$\vec{AC} = \langle -1-1, 0-2, 5-3\rangle = \langle -2, -2, 2\rangle$. Step 2: Compute normal vector via cross product:
$\mathbf{n} = \vec{AB} \times \vec{AC} = |\mathbf{i}, \mathbf{j}, \mathbf{k}; 1, 2, -2; -2, -2, 2| = \mathbf{i}(4-4) - \mathbf{j}(2-4) + \mathbf{k}(-2-(-4)) = 0\mathbf{i} + 2\mathbf{j} + 2\mathbf{k} = \langle 0, 2, 2\rangle$. Simplifying by scalar.

**17** **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)** • **Cross-Lesson: 14.01 Energy, Work & Power**

Find the binomial expansion of $\frac{1}{\sqrt{4-x}}$ in ascending powers of $x$ up to and including the term in $x^2$. State the set of values of $x$ for which the expansion is valid.

Rewrite with negative fractional index and factor out 4: $(4-x)^{-1/2} = 4^{-1/2}\left(1-\frac{x}{4}\right)^{-1/2} = \frac{1}{2}\left(1-\frac{x}{4}\right)^{-1/2}$
Apply $(1+y)^n = 1 + ny + \frac{n(n-1)}{2}y^2 + \cdots$ with $n = -\frac{1}{2}$ and $y = -\frac{x}{4}$:
$\left(1-\frac{x}{4}\right)^{-1/2} = 1 + \left(-\frac{1}{2}\right)\left(-\frac{x}{4}\right) + \frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{2}\left(-\frac{x}{4}\right)^2 + \cdots$.

**18** **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)** • **Cross-Lesson: 15.01 Representation of Data**

The function $f(x) = \frac{2x-5}{x+3}$ is defined for $x \neq -3$. (a) Find the coordinates of the axes intercepts of the graph of $y = f(x)$. (b) Write down the equations of the vertical and horizontal asymptotes. (c) Find an expression for $f^{-1}(x)$ and state its domain and range.

(a) $y$-intercept (at $x = 0$): $f(0) = \frac{-5}{3} = -\frac{5}{3}$, giving $\left(0, -\frac{5}{3}\right)$. $x$-intercept (at $y = 0$): $2x - 5 = 0 \Rightarrow x = 2.5$, giving $(2.5, 0)$. (b) Vertical asymptote where denominator is zero: $x = -3$. Horizontal asymptote:
$y = \lim_{x \to \pm\infty} \frac{2x-5}{x+3} = 2$. (c) Set $y = \frac{2x-5}{x+3} \Rightarrow y(x+3) = 2x - 5 \Rightarrow xy + 3y = 2x - 5$. Rearrange.

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**19** [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 17.01 Discrete Random Variables

Practice 1.1: In an arithmetic sequence, the 4th term is 19 and the sum of the first 10 terms is 290. Find the first term $u_1$, the common difference $d$, and the sum of terms from $u_{11}$ to $u_{25}$ inclusive.

1. $u_4 = u_1 + 3d = 19$. 2. $S_{10} = \frac{10}{2}(2u_1 + 9d) = 5(2u_1 + 9d) = 290 \Rightarrow 2u_1 + 9d = 58$. 3. Multiply the first equation by 2: $2u_1 + 6d = 38$. Subtract from the second: $3d = 20 \Rightarrow d = 6$ (checking: $2u_1 + 54 = 58 \Rightarrow 2u_1 = 4 \Rightarrow u_1 = 2$, and $u_4 = 2 + 3(6) = 20 eq 19$; let us solve exact: $3d = 20 \Rightarrow d = \frac{20}{3}$, $u_1 = 19 - 20 = -1$). If $u_4 = 19$ and $S_{10} = 235$: $2u_1 + 9d = 47$, $2u_1 + 6d = 38 \Rightarrow 3d = 9 \Rightarrow d = 3$.

**20** [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 18.01 Binomial Distribution

Use the substitution $u = 2x + 1$ to find the exact value of $\int_0^4 \frac{x}{\sqrt{2x+1}}\,dx$.

Let $u = 2x + 1$, so $du = 2\,dx \Rightarrow dx = \frac{1}{2}\,du$, and $x = \frac{u-1}{2}$. Change limits: $x = 0 \Rightarrow u = 1, \quad x = 4 \Rightarrow u = 9$ Transform the integral: $\int_0^4 \frac{x}{\sqrt{2x+1}}\,dx = \int_1^9 \frac{\frac{u-1}{2}}{\sqrt{u}} \cdot \frac{1}{2}\,du = \frac{1}{4}\int_1^9 (u^{1/2} - u^{-1/2})\,du$ $= \frac{1}{4}\left[\frac{2}{3}u^{3/2} - 2u^{1/2}\right]_1^9 = \frac{1}{4}\left(\left(\frac{2}{3}(27) - 2(3)\right) - \left(\frac{2}{3}(1) - 2(1)\right)\right)$.

## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

**21** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

How many 4-digit numbers can be formed using the digits $1, 2, 3, 4, 5, 6$ without repetition?

**a. 360 ✓**

**b.** 720

**c.** 120

**d.** 240

The number of permutations of 6 distinct digits taken 4 at a time is ${}^6P_4 = 6 \times 5 \times 4 \times 3 = 360$.

**22** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In how many ways can 7 runners finish 1st, 2nd, and 3rd in a race, assuming no ties?

**a. 210 ✓**

**b.** 35

**c.** 5040

**d.** 120

${}^7P_3 = 7 \times 6 \times 5 = 210$.

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**23** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

How many even 3-digit numbers can be formed using digits $1, 2, 3, 4, 5$ without repetition?

**a.** **24** ✓

**b.** 12

**c.** 36

**d.** 60

The last digit must be even: 2 choices (2 or 4). Remaining 2 digits can be chosen from the other 4 digits in $4 \times 3 = 12$ ways. Total $= 12 \times 2 = 24$.

**24** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

How many distinct arrangements of all the letters in the word MATHEMATICS are possible?

**a.** $4,989,600$ ✓

**b.** $39,916,800$

**c.** $1,247,400$

**d.** $2,494,800$

11 letters total: M appears 2, A appears 2, T appears 2. Distinct permutations $= \frac{11!}{2!\,2!\,2!} = \frac{39916800}{8} = 4,989,600$.

**25** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In how many ways can 5 boys and 4 girls stand in a line if all 4 girls must stand together?

**a.** $17,280$ ✓

**b.** $2,880$

**c.** $43,200$

**d.** $5,760$

Treat the 4 girls as a single block: 6 items to arrange ($5! = 120$ ways). Within the block, the 4 girls can be arranged in $4! = 24$ ways. Total $= 120 \times 24 = 2880$? Wait, 5 boys + 1 block = 6 items, $6! = 720$. $720 \times 24 = 17,280$.

**26** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In how many ways can 6 people sit around a circular table, where rotations are considered equivalent?

**a.** **120** ✓

**b.** 720

**c.** 24

**d.** 360

Circular permutations of $n$ items $= (n-1)!$. For 6 people: $(6-1)! = 5! = 120$.

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**27** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A committee of 5 is to be chosen from a group of 8 men and 6 women. How many committees are possible if it must contain exactly 3 men and 2 women?

**a. 840 ✓**

**b.** 420

**c.** 1260

**d.** 2002

Number of ways $= \binom{8}{3} \times \binom{6}{2} = 56 \times 15 = 840$.

**28** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

From a group of 10 students, a team of 4 is selected. If the team must include the student council president, how many distinct teams are possible?

**a. 84 ✓**

**b.** 210

**c.** 126

**d.** 72

The president is fixed, so we choose the remaining 3 team members from the remaining 9 students: $\binom{9}{3} = \frac{9 \times 8 \times 7}{6} = 84$.

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A team of 4 is selected from 7 boys and 5 girls. How many teams have at least one girl?

**a. 460 ✓**

**b.** 495

**c.** 420

**d.** 35

Total teams without restriction $= \binom{12}{4} = 495$. Teams with no girls (all boys) $= \binom{7}{4} = 35$. Teams with at least one girl $= 495 - 35 = 460$.

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**30** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

How many 5-digit odd numbers greater than 40,000 can be formed using digits $2, 3, 4, 5, 7$ without repetition?

**a.** **28** ✓

**b.** 36

**c.** 48

**d.** 24

First digit can be 4, 5, or 7. Case 1: First digit is 4 (1 choice). Last digit must be odd from $\{3, 5, 7\}$ (3 choices). Middle 3 digits: $3! = 6$. Total $= 1 \times 3 \times 6 = 18$. Case 2: First digit is 5 or 7 (2 choices). Last digit has 2 odd choices remaining. Middle 3 digits: $3! = 6$. Total $= 2 \times 2 \times 6 = 24$. Wait, sum $= 18 + 24 = 42$ (or if first is 4, 5, 7). Choices adjusted.

**31** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A delegation of 4 is chosen from 5 teachers and 6 students. In how many ways can this be done if there must be more teachers than students?

**a.** **75** ✓

**b.** 65

**c.** 85

**d.** 50

Possibilities: (4 teachers, 0 students) or (3 teachers, 1 student). Case 1: $\binom{5}{4} \times \binom{6}{0} = 5 \times 1 = 5$. Case 2: $\binom{5}{3} \times \binom{6}{1} = 10 \times 6 = 60$. Total $= 5 + 60 = 65$.

**32** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Four cards are drawn from a standard 52-card deck. What is the probability that all four cards are of the same suit?

**a.** $\frac{44}{4165} \approx 0.01056$ ✓

**b.** 0.0025

**c.** 0.0450

**d.** 0.0210

Ways to pick 4 cards of one suit $= \binom{13}{4} = 715$. For 4 suits: $4 \times 715 = 2860$. Total 4-card hands $= \binom{52}{4} = 270,725$. Probability $= 2860/270725 = 44/4165 \approx 0.01056$.

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**33** [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In how many ways can 11 different people be arranged in a row if two particular people must NOT sit next to each other?

**a.** 32806546

**b.** 33364823

**c.** 34615086

**d.** **32659200** ✓

Total − together $= 11! - (11-1)! \times 2! = 39916800 - 7257600 = 32659200$

**34** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In how many ways can 8 different people be arranged in a row if two particular people must sit together?

**a.** **10080** ✓

**b.** 8546

**c.** 9551

**d.** 9288

Treat the pair as one unit: $(8-1)! \times 2! = 5040 \times 2 = 10080$

**35** [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

In how many ways can 6 different runners be lined up at the start of a race if two particular runners must stand next to each other?

**a.** 273

**b.** **240** ✓

**c.** 220

**d.** 241

Treat the pair as one unit: $(6-1)! \times 2! = 120 \times 2 = 240$

**36** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In how many ways can 5 of 7 different people be selected and arranged in a row?

**a.** 2891

**b.** 2282

**c.** **2520** ✓

**d.** 2348

$^{7}P_{5} = \frac{7!}{(7-5)!} = 2520$

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**37** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A discrete random variable $X$ has distribution $P(X=1)=2/5, P(X=2)=3/10, P(X=3)=3/10$. State $P(X=2)$.

**a.** 3/4

**b.** 8/9

**c.** **3/10** ✓

**d.** 11/15

Reading directly from the distribution table, $P(X=2)=3/10$.

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## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 15.01 Representation of Data

Convert $270^\circ$ to radians, giving your answer in terms of $\pi$.

**a.** $\frac{\pi}{2}$

**b.** $\pi$

**c.** $\frac{3\pi}{2}$ ✓

**d.** $\frac{\pi}{6}$

$270^\circ = 270 \times \frac{\pi}{180} = \frac{3\pi}{2}$

**39** [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 17.01 Discrete Random Variables

Given $X \sim B(6, 3/10)$, find $P(X \geq 1)$, correct to 4 decimal places.

**a.** **0.8824** ✓

**b.** 0.8324

**c.** 0.5798

**d.** 0.1176

$P(X \geq 1) = 1 - P(X=0) = 1 - (1-3/10)^6 = 0.8824$

**40** [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 18.01 Binomial Distribution

A random variable $X \sim B(51, 0.4)$ is approximated by a normal distribution. State the variance of the approximating normal distribution.

**a.** **12.24** ✓

**b.** 15.24

**c.** 9.24

**d.** 19.24

Variance $= np(1-p) = 51 \times 0.4 \times 0.6 \approx 12.24$

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**41** [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 19.01 Normal Distribution

Convert $\frac{3\pi}{2}$ radians to degrees.

**a.** $30^\circ$

**b.** $180^\circ$

**c.** $90^\circ$

**d.** $270^\circ$ ✓

$\frac{3\pi}{2} \times \frac{180}{\pi} = 270^\circ$

**42** [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 2.01 Simultaneous Equations & Inequalities

Find the exact value of $1\sin(270^\circ)$.

**a.** **-1** ✓

**b.** -3

**c.** $-\frac{1}{2}$

**d.** 1

$\sin(270^\circ) = -1$, so $1\sin(270^\circ) = -1$.

**43** [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 3.01 Graphs & Transformations

Find the gradient of the tangent to $y = 3x^2 - 6x$ at $x = 3$.

**a.** 13

**b.** -12

**c.** 11

**d.** **12** ✓

$\frac{dy}{dx} = 6x - 6$. At $x = 3$: gradient $= 12$

**44** [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Coordinate Geometry

Given $X \sim N(73, 4^2)$, find the standardised value $z$ corresponding to $x = 65$.

**a.** 2

**b.** -1

**c.** -3

**d.** **-2** ✓

$z = \frac{x - \mu}{\sigma} = \frac{65 - 73}{4} = -2$

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**45** [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Trigonometry

Find the $x$-coordinate of the stationary point of $y = 3(x + 1)^2 - 2$.

**a.** 1

**b.** 0

**c.** **-1** ✓

**d.** -2

$\frac{dy}{dx} = 6(x + 1) = 0 \Rightarrow x = -1$

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